* Expansion of Liquids *

* liquids have no shape and size of their own. they aquire the shape of container.

So liquids have no linear & areal expansions but will have "volume expansion".

* liquids cannot heat alone. we can take the help of container to heat the liquids.

* Expansion of liquids is of two types.

1) Real or absolute expansion of liquid.

2) Apparent expansion of liquid.

Real Expansion of Liquid:

Expansion of "liquid alone" when it is heated is called real expansion of liquid.

Coefficient of real expansion of liquid (γ_r):

$$\gamma_r = \gamma_L = \frac{\text{real increase in volume of A liq}}{\text{or final volume} \times \text{rise in temp}}$$

$$\gamma_r = \frac{(\Delta V)_{real}}{V(t_2 - t_1)} \quad /^{o}c \ (or) \ /k$$

$\Rightarrow$ real expansion of liquid,

$$\boxed{\Delta V_{rel} = V \gamma_r (t_2 - t_1)}$$

* final real volume of liquid,

$$\boxed{V_f = V(1 + \gamma_r (t_2 - t_1))}$$

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* 'γ_r' - depends only on liquid but not on container.

Apparent Expansion of liquid:

* Expansion of liquid as seen with respect to container is called apparent expansion.

* It includes both expansion of liquid as well as vessel.

Coefficient of apparent expansion:

$$\gamma_a = \frac{\text{Apparent increase in volume}}{\text{original volume} \times \text{Rise in temp}}$$

$$\gamma_A = \frac{(\Delta V)_{app}}{V \times (t_2 - t_1)}$$

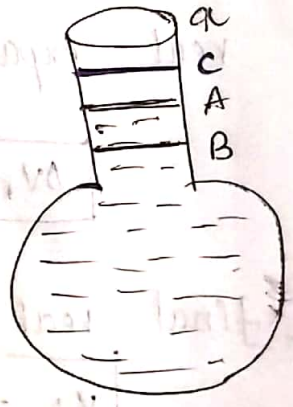
$\Rightarrow$ Apparent expansion,

$$\boxed{\Delta V_{app} = V \gamma_a (t_2 - t_1)}$$

* final apparent volume of liquid,

$$\boxed{V_f = V(1 + \gamma_a (t_2 - t_1))}$$

Relation b/w γ_a & γ_r:



* when the system is heated from $t_1^\circ C$ to $t_2^\circ C$,

Due to expansion of vessel, liquid level falls from A to B.

* now, due to expansion of liquid, level rises from B to C

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From fig,

$BC = AB + AC$

$a_1 BC = a_1 AB + a_1 AC$

⇒ real expansion of liquid = expansion of vessel + Apparent expansion of liquid

$$V\gamma_r(t_2 - t_1) = V\gamma_g(t_2 - t_1) + V\gamma_a(t_2 - t_1)$$

$$\Rightarrow \boxed{\gamma_r = \gamma_g + \gamma_a} \quad *$$

where $\boxed{\gamma_g = 3\alpha_g}$

we have,

$$\boxed{\gamma_a = \gamma_r - \gamma_g}$$

points:

* If $\gamma_r > \gamma_g \Rightarrow \gamma_a = +ve \Rightarrow \Delta V_{app} = +ve$

i.e, liq. level rises (w.r.to container) on heating.

* If $\gamma_r < \gamma_g \Rightarrow \gamma_a = -ve \Rightarrow \Delta V_{app} = -ve$

i.e, liq. level falls (w.r.to container) on heating.

* If $\gamma_g = -ve \Rightarrow \gamma_a = +ve \Rightarrow \Delta V_{app} = +ve$

i.e, liq level rises (w.r.to container) on heating.

* If $\gamma_g = 0 \Rightarrow \gamma_a = \gamma_r \Rightarrow \boxed{\Delta V_{app} = \Delta V_{real}}$

observed in all liquids except water.

* water has uneven rise of fall of liq. level.

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Variation of density of liquid with temp

At t°c, V_1 = vol of liq

ρ_1 = density of liq

$\therefore$ mass of liq, $m_1 = V_1 \rho_1$

when heated,

At t_2°c, V_2 = vol of liq

ρ_2 = density of liq

$\therefore$ mass of liq, $m_2 = V_2 \rho_2$

we have,

mass of liq will not change on heating / cooling.

i.e, $m_1 = m_2$

$$V_1 \rho_1 = V_2 \rho_2$$

we have, $V_1 \rho_1 = V_1 (1 + \gamma_r (t_2 - t_1)) \rho_2$

$V_2 = V_1 [1 + \gamma_r (t_2 - t_1)]$

$$\boxed{\rho_2 = \frac{\rho_1}{1 + \gamma_r (t_2 - t_1)}}$$

clearly, $V_1 \rho_1 = V_2 \rho_2$

As $V(\uparrow) \Rightarrow \rho(\downarrow)$

$\rho_2 + \rho_2 \gamma_r (t_2 - t_1) = \rho_1$

$\rho_2 \gamma_r (t_2 - t_1) = \rho_1 - \rho_2$

$$\boxed{\gamma_r = \frac{\rho_1 - \rho_2}{\rho_2 (t_2 - t_1)} \text{ per °c}}$$

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ρ_0 = density at 0°c

ρ_t = density at t°c

$$\rho_t = \frac{\rho_0}{1 + \gamma_r (t_2 - t_1)}$$

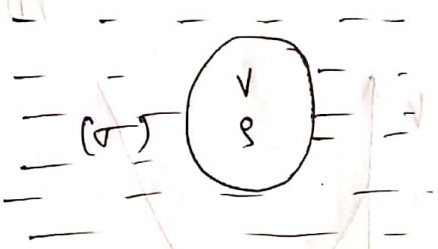
$$\boxed{\rho_t = \frac{\rho_0}{1 + \gamma_r t}}$$

(or) $$\rho_t = \rho_0 (1 + \gamma_r t)^{-1}$$

$$\boxed{\rho_t = \rho_0 (1 - \gamma_r t)} \quad **$$

Effect of thrust (Buoyancy) on the body due to heating:

Thrust on the body when it is completely submerged in liquid,



$$Th = V \sigma g$$

On heating,

new thrust, $Th' = V' \sigma' g$

$$Th' = \frac{V^0 (1 + \gamma_g (t_2 - t_1)) \sigma g}{[1 + \gamma_r (t_2 - t_1)]}$$

$$Th' = Th \left(\frac{1 + \gamma_g (t_2 - t_1)}{1 + \gamma_r (t_2 - t_1)} \right)$$

Since $\boxed{\gamma_r > \gamma_g}$ $\Rightarrow$ $\boxed{Th' < Th}$

$\therefore$ Thrust decreases on heating.

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Apparent wt of the body:-

$w_{app} = mg - Th$ (before heating)

$w'_{app} = mg - Th'$ (after heating)

Since $Th' < Th$

$\Rightarrow \boxed{w'_{app} > w_{app}}$

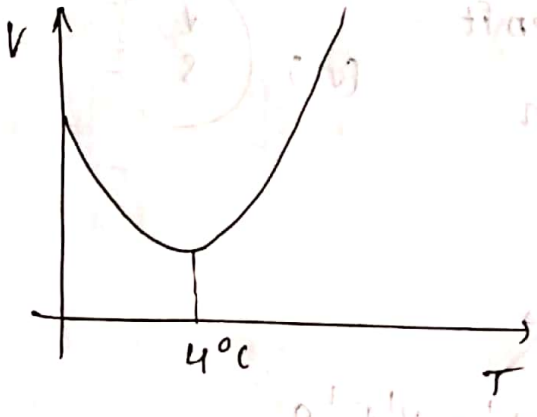
i.e, app wt of body increases on heating.

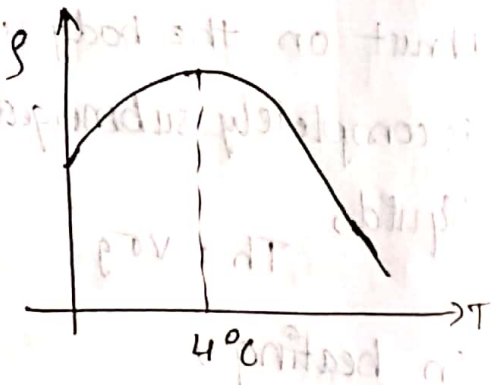
Anomalous expansion of water:-

$0^\circ C \xrightarrow{\text{heating}} 4^\circ C \xrightarrow{\text{heating}} 100^\circ C$

contracts | expansion

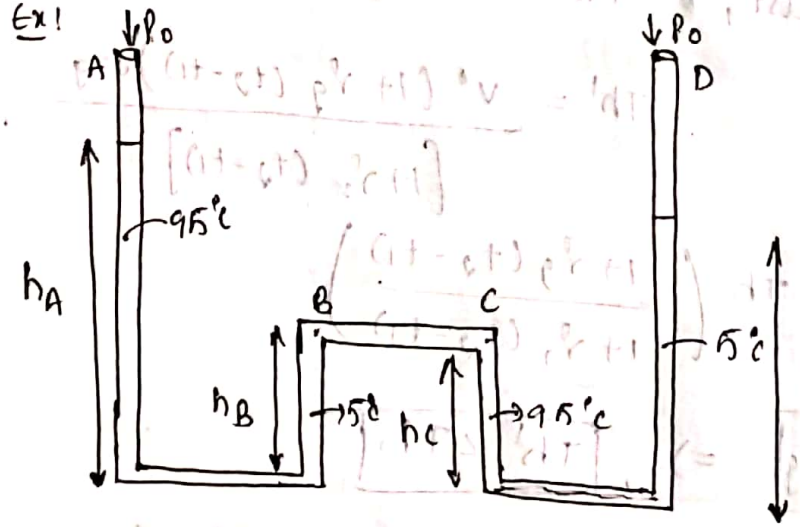
min volume
max density





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Ex!



Find ρ_ℓ of liquid?

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Sol: since B & C lies on the same line,

$$P_B = P_C$$

$$\cancel{P_0} + h_A \rho_{95} g - h_B \rho_5 g = \cancel{P_0} + h_B \rho_5 g - h_C \rho_{95} g$$

$$\rho_{95} = \frac{\rho_5}{1+\gamma_r(95-5)}$$

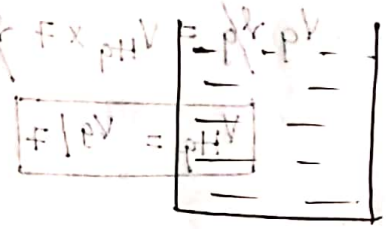
$$\rho_{95} = \frac{\rho_5}{1+90\gamma_r}$$

$$\rho_{95} + 90\rho_{95}\gamma_r = \rho_5 \Rightarrow 90\rho_{95}\gamma_r = \rho_5 - \rho_{95}$$

$$\boxed{\gamma_r = \frac{\rho_5 - \rho_{95}}{90\rho_{95}}}$$

Applications :-

* liquid completely fills the beaker at t_1°c, when it is heated to t_2°c.



Volume expansion of vessel,

$$\Delta V_1 = V \gamma_g (t_2 - t_1)$$

Volume expansion of liquid,

$$\Delta V_2 = V \gamma_l (t_2 - t_1)$$

Volume overflow on heating,

$$\Delta V = \Delta V_2 - \Delta V_1$$

$$\boxed{\Delta V = V(\gamma_l - \gamma_g)(t_2 - t_1)}$$

* Condition for ~~one~~ occupied volume of liquid remains same at all temperatures.



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$V_c = V_l + V_{air}$ (before heating)

$V_c' = V_l' + V_{air}$ (after heating)

$V_c' - V_c = V_l' - V_l$

$\Delta V_c = \Delta V_l$

$V_c \gamma_c \Delta t = V_l \gamma_l \Delta t$

$$\boxed{V_c \gamma_c = V_l \gamma_l} *$$

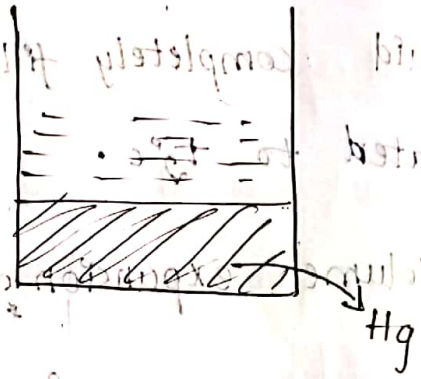
we have,

$$\boxed{\gamma_{Hg} = 7 \gamma_{glass}} *$$

Here, $V_g \gamma_g = V_{Hg} \gamma_{Hg}$

$V_g \gamma_g = V_{Hg} \times 7 \gamma_g$

$$\boxed{V_{Hg} = V_g / 7}$$



Dilatometer:

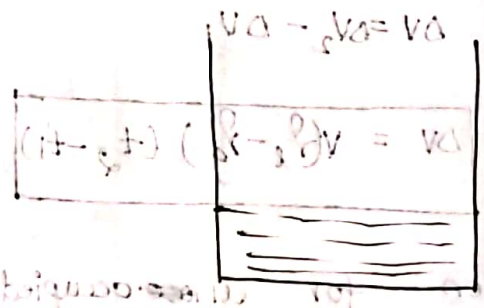
Here, Expansion of water w.r.to container

= Real expansion of water

* Condition for fractional volume of liquid is same at all temperatures.

i.e, $\frac{V_l}{V_t} = \text{const}$

i.e, $\frac{V_l}{V_c} = \frac{V_l'}{V_c'}$



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$$\frac{V_l}{V_c} = \frac{V_l(1+\gamma_l \Delta t)}{V_c(1+\gamma_g \Delta t)}$$

$$1+\gamma_l \Delta t = 1+\gamma_g \Delta t$$

$$\boxed{\gamma_l = \gamma_g = 3\alpha_g}$$

* condition for fractional volume of air is same at all temps.

$$\frac{V_{air}}{V_c} = \text{const}$$

$$\frac{V_c - V_l}{V_c} = \text{const}$$

$$\frac{V_c - V_l}{V_c} = \frac{V_c' - V_l'}{V_c'}$$

$$\frac{V_l}{V_c} = \frac{V_l'}{V_c'}$$

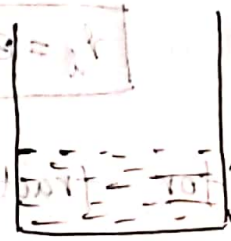
$$\boxed{\gamma_l = \gamma_g = 3\alpha_g}$$

* condition for height of liquid w.r.to ground (C.O.M of liq) remains same at all temps.

we have,

height of liquid,

$$h = \frac{V_l}{A_c}$$



$$\frac{V_l}{A_c} = \frac{V_l'}{A_c'}$$

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$$\frac{V_l}{A_l} = \frac{V(1+\gamma_l \Delta t)}{A(1+\beta \Delta t)}$$

$$1+\gamma_l \Delta t = 1+\beta \Delta t$$

$$\boxed{\gamma_l = \beta = 2\alpha g}$$

For partially submerged bodies:

Consider a cube of side length l



① Condⁿ for depth (h) of block inside the liquid remains same at all temp.

we have,

$$mg = F_B$$

$$mg = l^2 h d_l g \quad \text{(before heating)}$$

& $$mg = l_1^2 h d_l' g \quad \text{(after heating)}$$

$$l^2 h d_l g = l_1^2 h d_l' g$$

$$A d_l = A_1 d_l'$$

$$A d_l = A(1+\beta \Delta t)\frac{d_l}{1+\gamma_l \Delta t}$$

$$\boxed{\gamma_l = \beta = 2\alpha g}$$

② Condⁿ for fractional submerged volume of body remains same at all temps.

we have,

$$mg = F_B$$

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$$V d_B g = V_{sub}\, d_l\, g$$

$$\frac{V_{sub}}{V} = \frac{d_B}{d_l} = \text{const}$$

$$\frac{d_B}{d_l} = \frac{d_B'}{d_l'}$$

$$\frac{d_B}{d_l} = \frac{d_B}{1+\gamma_B \Delta t} \times \frac{1+\gamma_l \Delta t}{d_l}$$

$$\boxed{\gamma_B = \gamma_l = 3\alpha_B} \quad *$$

③ cond^n for body just to sink into the liquid.

here, $d_B' = d_l'$

$$\frac{d_B}{1+\gamma_s (t_2 - t_1)} = \frac{d_l}{1+\gamma_l (t_2 - t_1)}$$

* If expansion of solid is neglected

$$\Rightarrow \boxed{\gamma_s = 0}$$

$$\boxed{d_B = \frac{d_l}{1+\gamma_l (t_2 - t_1)}} \quad *$$

Again we have,

$$mg = F_B.$$

$$\Rightarrow V d_s g = V_{sub}\, d_l\, g$$

fractional volume of body submerged in liquid

$$f_s = \frac{V_{sub}}{V} = \frac{d_s}{d_l} \quad \text{(before heating)}$$

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and $f_s' = \frac{d_s'}{d_l'}$ (after heating)

$$\frac{f_s'}{f_s} = \frac{d_s'}{d_l'} \times \frac{d_l}{d_s}$$

$$= \frac{d_s}{1+\gamma_s \Delta t} \times \frac{1+\gamma_l \Delta t}{d_l} \times \frac{d_l}{d_s}$$

$$\boxed{\frac{f_s'}{f_s} = \frac{1+\gamma_l \Delta t}{1+\gamma_s \Delta t}} \quad \text{---} (a)$$

① If $\gamma_l > \gamma_s$ then $f_s' > f_s$

body sinks further on heating.

② If $\gamma_l < \gamma_s$ then $f_s' < f_s$

body lifts up on heating.

③ If $\gamma_l = \gamma_s$, $f_s = f_s'$

no effect on submergence.

If expansion is neglected

$\gamma_s = 0$

then $\frac{f_s'}{f_s} = 1+\gamma_l \Delta t$

$$\boxed{\gamma_l = \frac{f_s' - f_s}{f_s \Delta t}}$$

and $f_s' = f_s(1+\gamma_l \Delta t)$

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Here, $f_{s_1} = f_{s_0}(1+\gamma_d \Delta t_1)$ $\quad f_{s_0}$ = submerged fraction

$f_{s_2} = f_{s_0}(1+\gamma_d \Delta t_2)$ $\quad$ at σ_c

$$\frac{f_{s_1}}{f_{s_2}} = \frac{1+\gamma_d t_1}{1+\gamma_d t_2}$$

$$f_{s_1} + f_{s_1}\gamma_d t_2 = f_{s_2} + f_{s_2}\gamma_d t_1$$

$$\boxed{\gamma_d = \frac{f_{s_2} - f_{s_1}}{f_{s_1} t_2 - f_{s_2} t_1}}$$

Again eqn @ becomes,

$$\frac{f_s'}{f_s} = \frac{1+\gamma_d \Delta t}{1+\gamma_s \Delta t}$$

$$\frac{f_s'}{f_s} - 1 = \frac{1+\gamma_d \Delta t}{1+\gamma_s \Delta t} - 1$$

$$\frac{f_s' - f_s}{f_s} = \frac{1+\gamma_d \Delta t - 1 - \gamma_s \Delta t}{1+\gamma_s \Delta t}$$

$$\frac{f_s' - f_s}{f_s} = \frac{(\gamma_d - \gamma_s)\Delta t}{1+\gamma_s \Delta t}$$

$$\frac{\Delta f_s}{f_s} = (\gamma_d - \gamma_s)\Delta t \; (1 - \gamma_s \Delta t)$$

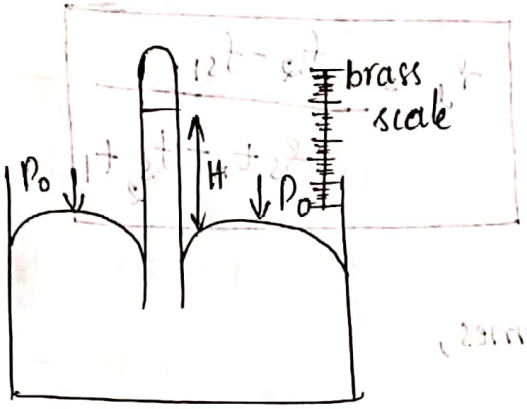
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$$= (\gamma_l - \gamma_s)\Delta t - (\cancel{\gamma_l - \gamma_s})\gamma_s \Delta t^2$$

$$\boxed{\frac{\Delta f_s}{f_s} = (\gamma_l - \gamma_s)\Delta t}$$

↳ fraction change in vol (submerged volume) of body on heating.

Correction for barometer readings



Brass scale is calibrated at $0^\circ C$.

It gives correct reading at $0^\circ C$

If H = measured reading at $t^\circ C$

then its true reading

$$H_t = H\alpha(t-0)$$

Since atmospheric pressure is same at all temps.

$$H_0 s_0 f = H_t s_t f$$

$$H_0 \rho_0 = H(1+\alpha_{scale}\Delta t)\frac{\rho_0}{1+\gamma_{Hg}\Delta t}$$

$$H_0 = H[1 - \gamma_{Hg}t + \alpha_{scale}\Delta t - \cancel{\alpha_{scale}}\,\alpha_{Hg}\Delta t^2]$$

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$$H_0 = H[1 - (\gamma_{Hg} - \gamma_{scale})\Delta t]$$

$$\Delta t = t - 0 = t$$

$$\boxed{H_0 = H[1 - (\gamma_{Hg} - \alpha_{scale})t]}$$

H_0 → true value at 0°C

H → measured value at t°C